Python Basics – Day 1

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1. What is a Variable?

A variable is a container used to store a value.

Example:

a = 10
name = "Rama"

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2. Rules for Declaring Variables

1. Must not start with a number
   ❌ 1a = 10
   ✅ a1 = 10
2. Must not start with special characters (except underscore _)
   ❌ @a = 5
   ✅ _a = 5
3. Must not be a reserved keyword
   ❌ int = 10
   ❌ float = 5
4. Should contain only alphabets, numbers, underscore

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3. Valid & Invalid Variables

Variable Name	Valid / Invalid	Reason
a = 10	Valid	Starts with alphabet
_i = True	Valid	Starts with underscore
-x = 5	Invalid	Special character
2num = 10	Invalid	Starts with number

name = Rama	Invalid	String not in quotes
name = "Rama"	Valid	String in quotes

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4. Data Types

Definition:
A data type defines what kind of value a variable stores.

Types of Data Structures

Primitive (stores single value):

- int
- float
- str
- bool

Non-Primitive (stores multiple values):

- list
- tuple
- set
- dictionary

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5. Primitive vs Non-Primitive

Primitive: Stores one value
Examples: 10, 2.5, "Hi", True

Non-Primitive: Stores multiple values
Examples: [1,2,3], (10,20), {"a":10}

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6. Linear vs Non-Linear Data Structures

Linear Data Structures:

- Array

- Stack
- Queue
- List

Non-Linear Data Structures:

- Tree
- Graph

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7. Built-in vs User-defined Structures

Built-in Data Structures:

- list
- tuple
- set
- dictionary

User-defined Data Structures:

- Stack
- Queue
- Linked List
- Graph
- HashMap

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8. Dynamically Typed Language

Python is dynamically typed, meaning there is no need to declare data types explicitly.

Example:

a = 10 # int
b = 2.5 # float
name = "Raj" # string

Definition:
Python automatically detects the data type at runtime.

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9. Print Statement

The print() function is used to display output or extract variable values.

Example:

```
print("Hello")
print(a)
print(a + b)
```

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10. Indentation

Python uses indentation instead of braces.

All statements must start from the left margin unless inside a block.

Example:

```
x = 10
if x > 5:
print("Greater")
```

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11. Interpreter

Python is an interpreter-based language.

It executes code line by line and stops at the first error.

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12. Simple Interest Program

```
P = 1000 # Principal
T = 1 # Time (years)
R = 10 # Rate of Interest

SI = (P * T * R) / 100

print("Simple Interest:", SI)
```

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13. ER Diagram Requirement (Student Management System)

Entities:

Student

- student_id (PK)
- name
- age
- city

Course

- course_id (PK)
- course_name
- duration

Enrollment

- enrollment_id (PK)
- student_id (FK)
- course_id (FK)
- enroll_date

Relationships:

A Student can enroll in many Courses.
A Course can have many Students.
Relationship type: Many-to-Many (resolved using Enrollment table).

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14. Assignment Questions

Part A – Theory

1. What is a variable?
2. Rules for declaring variables.
3. What is a data type?
4. Primitive vs non-primitive.
5. What is a dynamically typed language?
6. Features of Python.
7. What is programming?

8. What is Python?
9. What is Python community?
10. Difference between compiler and interpreter.
11. What is indentation?
12. What is print statement?